

Evolution Outperforms Random Search but Not Cost-Complexity Pruning: A Pre-Registered Test of Multi-Objective Evolutionary Decision-Tree Induction

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ABSTRACT

Multi-objective evolutionary algorithms have been proposed for inducing decision trees that trade accuracy against size, and the system studied here was previously reported to find smaller trees at “equivalent” accuracy to CART. We tested that claim under a pre-registered protocol with binding kill criteria: 20 OpenML-CC18 datasets, nested cross-validation, hypervolume over (test accuracy, node count), a random-search baseline matched to the GA’s evaluation count exactly, and CART’s full cost-complexity pruning path as the comparator. Two of the three hypotheses were rejected. NSGA-II recovers a better frontier than budget-matched random sampling of the same tree space (mean $\Delta HV +0.66$, Holm-adjusted $p = 0.032$, 15/20 datasets), so the evolutionary machinery contributes. But it has the larger hypervolume than CART’s pruning path on only 45% of datasets, below the pre-registered 60% threshold; and its tuned single tree loses more than 2 accuracy points to tuned CART on 8 of 20 datasets, so the pre-registered equivalence criterion fires. The losses concentrate on datasets where accuracy keeps rising with tree size: the GA’s delivered frontiers stop at a median of 5.4 nodes against CART’s 32.0. We also report what the audit behind this study found: previously published figures for this system that no run had produced, and two harness errors, in opposite directions, each of which reversed a kill-criterion verdict. We argue that budget-matched random search and full pruning paths should be the default comparators for evolutionary tree induction.

KEYWORDS

decision trees, evolutionary multi-objective optimisation, NSGA-II, interpretability, benchmarking, pre-registration, negative results

1 INTRODUCTION

Greedy recursive partitioning [7] optimises a split criterion that decomposes over nodes and controls size afterwards by pruning. Evolutionary tree induction [2, 20, 29] instead scores whole trees, and with a multi-objective algorithm such as NSGA-II [9] it returns a set of trees along the accuracy–size trade-off from one run [34]. The standard argument for it is that a global search over whole trees should find models that greedy induction misses: as small, or smaller, at the same accuracy. The comparisons surveyed by Barros et al. use greedy inducers such as C4.5 and CART as baselines, mostly on UCI datasets, and many report similar predictive performance with smaller trees [2].

Testing that argument requires two further baselines, neither of which appears among the comparators that survey lists. The first is *random search over the same tree space with the same number of fitness evaluations* [3]: without it, a favourable result cannot be

attributed to evolution rather than to the representation, the fitness or the sampling distribution. The second is CART’s *entire* cost-complexity pruning path rather than one tuned or depth-limited tree: a single CART tree is one point on a trade-off, and comparing a small evolved tree with it on accuracy compares two points at different sizes rather than two methods.

This paper reports a test of one such system—ours—against both baselines. It is also, in part, a negative-results paper, and its origin bears on how it should be read. The project’s public materials claimed “46–82% smaller trees with statistically equivalent accuracy”. An audit of every public claim against the code found that the headline figures had been typed into a plotting script as literals, that the remaining ones were stale, that the “equivalence” rested on non-significant cross-fold t -tests (a non-significant difference is not evidence of equivalence, and the tests treated non-independent folds as independent [12]), and that on the newest real run the GA lost to depth-matched CART on 7 of 8 datasets. We withdrew the claims, fixed three hypotheses and four kill criteria in writing before re-running anything, and report the outcome whichever way it fell [26].

Contributions.

- A pre-registered, budget-matched test of NSGA-II decision-tree induction on 20 OpenML-CC18 datasets. Evolution beats random sampling of the same space (H3 holds); it does not beat CART’s pruning path (H1 rejected); and it does not reach equivalent accuracy to tuned CART (H2 rejected).
- A mechanism for the H1 failure: the GA’s frontiers are truncated at small trees, and its hypervolume deficit against CART tracks the accuracy that only larger trees reach (Spearman $\rho = 0.85$, exploratory).
- An account of two evaluation-harness errors—selection on the test fold, and a per-fold hypervolume reference point—that each flipped a kill-criterion verdict, in opposite directions. Neither is visible in the headline statistics alone.
- Sensitivity and exploratory analyses (constraint repair; the optimal sparse-tree solver GOSDT [21]) and an open, reproducible harness in which every reported number is generated from committed run output.

2 RELATED WORK

Evolutionary tree induction. Barros et al. [2] survey evolutionary algorithms that induce whole decision trees or design their components; Kretowski [20] presents a framework for global evolutionary induction, Rivera-Lopez et al. [29] review metaheuristic

approaches more broadly, and *evtree* [17] is an R implementation of single-objective global evolutionary tree induction. Multi-objective variants optimise accuracy against size with Pareto selection [34]. Freitas [15] argued that weighted formulas, which fix the trade-off between criteria in advance, are largely ad hoc, and that Pareto and lexicographic approaches are more principled. In the comparisons surveyed by Barros et al., the baselines are greedy inducers and the datasets mostly come from UCI [2]; none of the listed comparators isolates the contribution of the search operators from that of the representation, which is the gap our random-search baseline targets.

Optimal sparse trees. Exact methods find provably optimal trees under a sparsity penalty or depth limit: OCT [4] via mixed-integer programming, and OSDT [19], GOSDT [21], DL8.5 [1] and MurTree [10] via branch-and-bound and dynamic programming over binarised features, made practical on continuous data by threshold guessing [24]. Sweeping the penalty yields a frontier directly comparable to ours. The case for evolution against them is flexibility, not optimality: arbitrary, non-decomposable objectives and an anytime answer.

Evaluation methodology. We follow Demšar [11] in treating datasets as the unit of inference, with Wilcoxon signed-rank tests and Holm correction [18]; nested cross-validation to avoid selection bias [8, 33]; TOST for equivalence [31]; and the Nadeau–Bengio variance correction where folds must be the unit [25]. Pre-registration [26] and variance-aware benchmarking [6] respond to the same problems we found in our own earlier claims [23].

Measuring interpretability. There is no agreed measure of how interpretable a tree is [13, 22]. Size- and path-length-based proxies are the most common [16, 28]; we report only those, and do not claim that any of them is interpretability [30].

3 METHOD

3.1 Representation and operators

Individuals are binary axis-parallel trees with a maximum depth of 6 and minimum sample counts per split (8) and per leaf (3). Leaves predict the majority class of the training rows reaching them. Initial trees grow recursively, stopping at each node with probability 0.3. Split thresholds are drawn from the midpoints between consecutive distinct values of the samples reaching the node—CART’s candidate set—filtered so both sides meet the leaf minimum. Variation uses subtree crossover and four mutations: threshold perturbation snapped to an observed midpoint, feature replacement, subtree pruning and leaf expansion (weights 0.45/0.25/0.25/0.05).

3.2 Fitness and selection

Each training fold is split 80/20 (stratified) into a fitting part and a validation part. Leaves are fitted on the former and the tree is scored on the latter, so the search does not reward trees for memorising the rows their leaves were fitted on. The multi-objective search uses NSGA-II [9] (DEAP implementation [14]) over (validation accuracy, –node count), population 80, 40 generations, crossover 0.72, mutation 0.18. Structurally identical individuals are removed before environmental selection: crowding distance

does not separate clones, and in a preliminary run without this step the number of distinct objective vectors fell from 26 to 2 within five generations. The delivered model set is the final population’s first front. The point-estimate run (Section 5.3) instead uses a single-objective GA with the same representation and operators, elitism of 12%, and a weighted sum of validation accuracy and the composite “interpretability” score named in K4, which is a search heuristic and not an outcome; its accuracy weight is tuned by inner CV and it delivers its best individual.

3.3 Baselines

Random search (RS). Draws trees from the GA’s own initialiser, scores them with the GA’s own fitness and validation split, and keeps a non-dominated archive on those validation objectives. Its budget is set per fold to the number of evaluations the GA actually spent on that fold, read from a counter rather than predicted: NSGA-II’s cost per generation depends on how many offspring variation changes, so the nominal $pop \times gens$ (3,200) overstates it. For the single-objective GA, elites carry their fitness across generations, so a run costs $pop + gens \times (pop - n_{elite}) = 2,920$ evaluations. **Archived GA.** Because RS can only deliver an archive, the GA was also run delivering its archive over every candidate it scored, as a control for that bookkeeping difference. **CART ccp path.** Scikit-learn CART [27] with the same sample minimums, fitted at up to 12 values spread along its cost-complexity pruning path—a frontier obtained from one greedy fit, with no search over whole trees. As pre-registered, the path is not depth-limited, whereas the GA cannot exceed depth 6; Section 5.2 also reports a depth-capped path.

3.4 Frontier metric

Every delivered model is scored once on the held-out test fold; each method’s point set is filtered for dominance in (test accuracy, node count), and its hypervolume is the area it dominates up to a reference point of accuracy 0 and one node more than the largest tree any method produced on that dataset [35]. The reference is fixed per dataset, not per fold (Section 6). Dominance filtering after scoring does not change the hypervolume, since dominated points add no area; every method delivers a set chosen without test labels. For display we divide hypervolume by the reference node count, giving an area fraction comparable across datasets; all tests use the raw values.

4 PRE-REGISTERED PROTOCOL

The hypotheses, datasets, statistics and kill criteria were written down before any re-run; deviations were recorded, with their expected direction, before the results they affect were computed.

H1 The GA frontier dominates CART’s pruning-path frontier by hypervolume on a majority of datasets.

H2 Held-out accuracy of the GA’s selected tree is equivalent to inner-CV-tuned CART within ± 2 accuracy points (TOST).

H3 The GA’s hypervolume exceeds budget-matched random search’s.

Kill criteria. K1: if RS matches the GA (Wilcoxon over datasets, $\alpha = 0.05$), the evolutionary machinery contributes nothing; publish the software only. K2: if the GA has the larger hypervolume

than CART on fewer than 60% of datasets, H1 is rejected, and may not be softened to “competitive on some datasets”. K3: if the GA loses more than 2 points to tuned CART on more than 30% of datasets, no equivalence claim is made. K4: the composite “interpretability” score used inside the fitness may not appear as an outcome.

Datasets. All OpenML-CC18 [5, 32] classification tasks with 500–1500 rows, at most about 50 features and a smallest class of at least 40: exactly 20 datasets, fixed by rule before any run reported here. The rule excludes *iris* and *wine* (fewer than 500 rows), on which most of the withdrawn claims rested; the Wisconsin diagnostic breast-cancer data, the third, is included as *wdbc* because it meets the rule.

Evaluation. Outer 10×3 repeated stratified CV for the frontier run (30 folds per dataset); per-fold seeds derived from (dataset, fold, method). The point-estimate run (H2/K3) tunes every method by inner 5-fold CV—CART over depth and α on its pruning path, the GA and RS over the accuracy weighting $\{0.5, 0.7, 0.9\}$ —and uses 10×1 outer CV. Both the single outer repeat and the removal of depth $\{4, 6, 8\}$ from the GA’s and RS’s grid are recorded deviations for compute; the latter handicaps the GA relative to CART, which is the conservative direction for a criterion that exists to block an equivalence claim. Complexity is reported as nodes, leaves, mean decision-path length (over training samples) and distinct features used.

Deviations. Recorded before use: (i) midpoint thresholds, for both searchers; it was measured beforehand to help RS more than the GA, i.e. to make K1 harder; (ii) validation-split fitness, for both searchers; (iii) the corrected budget formula above; (iv) 10×1 outer CV and a weight-only GA/RS grid for K3; (v) a loader fix for a pandas-version change that re-encoded two datasets, after which all 20 datasets reproduce the committed CART frontiers exactly; (vi) constraint repair (Section 5.4), written after K1/K2 were known and therefore reported only as sensitivity.

5 RESULTS

5.1 K1: NSGA-II versus budget-matched random search

Budgets matched on all folds (0 unequal; mean 2,546 evaluations per fit). The GA’s hypervolume exceeds budget-matched RS’s by +0.66 on average ($p = 0.011$, Holm $p = 0.032$, $d_z = +0.50$), on 15 of 20 datasets (Figure 1). K1 is not triggered. The archived-GA control does not differ significantly from the standard GA (+0.12, Holm $p = 0.286$), so there is no evidence that the advantage is an artefact of delivering a final population rather than an archive. The four-method Friedman test is not significant ($p = 0.202$; mean ranks GA 2.15, CART 2.25, archived GA 2.70, RS 2.90), so no Nemenyi post-hoc is licensed; K1 is defined on the pairwise test, which has more power here because two of the four methods are near-duplicates.

5.2 K2: NSGA-II versus the pruning path

The GA has the larger mean hypervolume than CART’s pruning path on 9 of 20 datasets (45%), below the 60% threshold: **H1 is rejected**. The mean difference is -4.21 (Holm $p = 0.368$). The depth asymmetry does not explain this: against CART’s path re-derived with the GA’s depth cap on the same folds and seeds (exploratory),

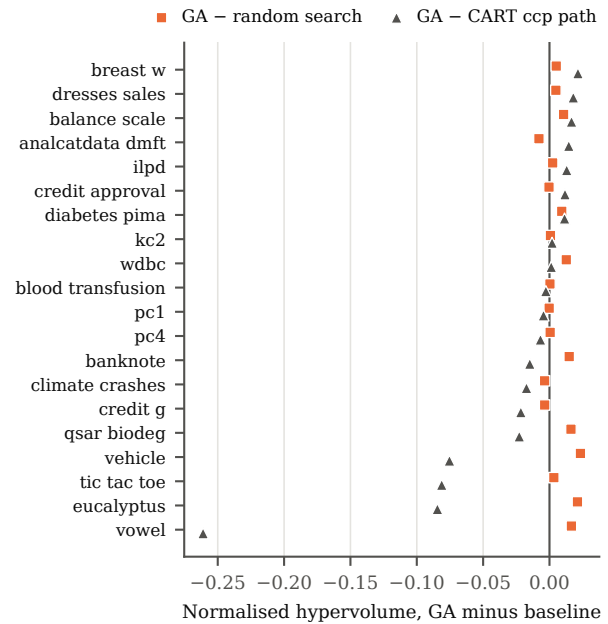


Figure 1: Per-dataset difference in normalised hypervolume between the GA and each baseline (mean over 30 outer folds). Right of zero favours the GA. Datasets sorted by the CART difference.

the GA wins on 8 of 20 datasets (40%). Table 1 and Figure 1 show the pattern: the GA wins narrowly on the datasets where small trees are near-optimal, and loses heavily on *vowel*, *eucalyptus*, *tic-tac-toe* and *vehicle*, where accuracy keeps rising with size.

Exploratory, post hoc. The GA’s delivered frontiers are truncated. The largest tree on its test front has a median of 5.4 nodes across datasets, against 11.2 for RS and 32.0 for CART, and its best test accuracy averages 0.750 against CART’s 0.801. Across datasets the hypervolume deficit to CART is strongly rank-correlated with that best-accuracy gap (Spearman $\rho = 0.85$, $p < 0.001$). Figure 2 shows one fold of each kind: in both, the GA’s front lies at or above CART’s over the size range it covers, and does not extend to the large trees.

Exploratory ablation. To test candidate causes we changed one thing at a time on the first 10 folds of the 4 datasets where the GA loses most, with the committed GA as a control arm that reproduces the original fronts exactly. Scoring fitness on the fitting data instead of a validation split (so that a larger tree need not prove itself on about a hundred held-out rows) closes 29% of the normalised-hypervolume gap to CART on average. Doubling population and generations closes 28%. Removing the bias toward small trees—seeding full-depth trees and swapping the expand and prune mutation weights—closes -4% , i.e. none of it. Under every variant the largest delivered tree stays between 15 and 40 nodes against CART’s 78–162. On these datasets the truncation is therefore not an initialisation artefact; held-out selection and

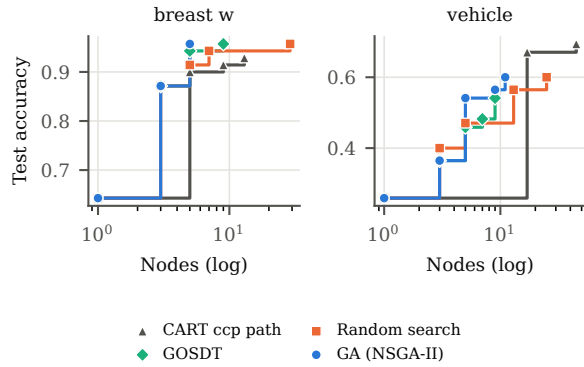


Figure 2: Test-fold frontiers for outer fold 1 of a dataset the GA wins (*breast-w*) and one it loses (*vehicle*).

Table 1: Mean normalised hypervolume per dataset (higher is better; best of GA, RS and CART in bold) and the mean node count of the largest tree on the GA’s and CART’s test fronts. GOSDT (Section 5.5) was run on the first 10 folds with its own reference point, so its column is not on the same scale and is not bolded.

Dataset	GA	RS	CART	GOSDT	Largest tree	
					GA	CART
analcadata_dmft	0.237	0.245	0.223	0.228	6	94
balance_scale	0.749	0.738	0.732	0.704	13	39
banknote	0.894	0.879	0.909	0.890	13	29
blood_transfusion	0.782	0.781	0.784	0.775	6	32
breast_w	0.906	0.901	0.884	0.861	5	13
climate_crashes	0.892	0.896	0.910	0.895	2	10
credit_approval	0.840	0.840	0.828	0.780	4	22
credit_g	0.723	0.727	0.744	0.729	5	48
diabetes_pima	0.744	0.734	0.732	0.758	6	32
dresses_sales	0.620	0.615	0.602	0.602	5	36
eucalyptus	0.521	0.500	0.606	0.509	12	63
ilpd	0.723	0.721	0.710	0.709	4	13
kc2	0.827	0.826	0.825	0.838	3	13
pc1	0.919	0.919	0.924	0.921	2	16
pc4	0.887	0.886	0.893	0.892	4	23
qsar_biodeg	0.789	0.773	0.812	–	9	45
tic_tac_toe	0.744	0.740	0.825	0.800	10	118
vehicle	0.594	0.570	0.669	0.603	14	77
vowel	0.360	0.343	0.621	0.399	24	155
wdbc	0.899	0.886	0.898	0.829	5	11

search budget each account for part of it, and most of it is left unexplained by these three factors. By contrast, CART’s path reaches its large trees without scoring them on held-out data.

5.3 K3 and H2: accuracy of a single tuned tree

With its accuracy weighting tuned by inner CV, the single-objective GA’s tree loses more than 2 points to tuned CART on 8 of 20 datasets, above the 30% threshold: K3 fires under the primary reading. The corrected per-dataset TOST fails to establish equivalence on 18 of 20, so the secondary reading agrees. Across datasets the mean difference is -0.038 (90% CI $[-0.069, -0.008]$, TOST $p = 0.843$): the accuracies are not equivalent within ± 2 points,

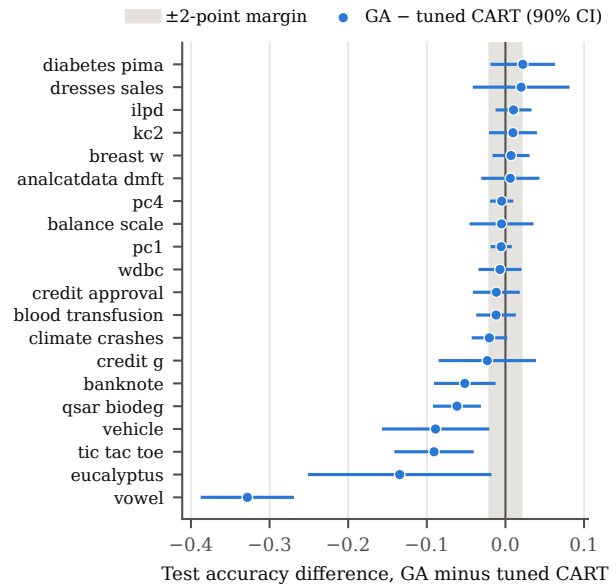


Figure 3: Test-accuracy difference between the tuned GA tree and tuned CART per dataset, with 90% intervals using the Nadeau–Bengio corrected variance over outer folds. The shaded band is the pre-registered ± 2 -point margin.

H2 is rejected, and no equivalence claim is made. The difference is not significant either (Wilcoxon, Holm $p = 0.072$). This combination—a difference that is neither significant nor shown to be within the margin—is the one the earlier “equivalence” claim misread as evidence of equivalence.

The losses are not spread evenly. Of the 8 datasets where the GA loses, 8 are among the 8 with the largest hypervolume deficit to CART in the frontier run—a separate run in a different algorithm mode (Figure 3); on the other 12 the GA is within 2 points of CART or ahead of it (ahead by more than 2 points on 1). Mean accuracies are GA 0.738, tuned CART 0.776, RS 0.726. The GA’s trees have 6.3 leaves (CART 18.4), a mean decision path of 2.24 splits (CART 3.61) and use 2.9 features (CART 6.3). Inner CV chose the most accuracy-favouring weight on offer, 0.9, in 52% of folds, and the trees remain small at that weight. Because the upper end of the grid was chosen in about half the folds, we cannot exclude that a higher accuracy weight would yield larger and more accurate trees. As in K1, the GA also beats budget-matched RS on accuracy ($+0.012$, Holm $p = 0.003$).

5.4 Sensitivity: constraint repair

The sample-count constraints were enforced only at initialisation. On one fold of each of 4 datasets, 8–27% of the trees NSGA-II evaluated contained a split the constraints forbid (0% for RS, whose trees all come from the initialiser), yet only 1 of 79 internal nodes in the delivered fronts did, consistent with the size objective removing such splits on its own. We added a repair that collapses, top-down, every split reached by fewer than 8 training rows or leaving fewer

than 3 on a side. Re-running the frontier benchmark with repair on and everything else fixed gives a GA–RS difference of +0.85 (Holm $p = 0.034$, 16/20 datasets), and the GA has the larger hypervolume than CART on 45% of datasets. Neither verdict changes. One secondary result does: with repair, the standard GA also beats its archived variant (+0.54, Holm $p = 0.025$), so the choice between delivering the final front and the archive is not neutral once dead splits are removed.

5.5 Exploratory: GOSDT

GOSDT [21] with threshold guessing [24], swept over ten regularisation values at the same depth limit, run on the first 10 outer folds and scored against the same per-dataset reference, has the larger hypervolume than CART’s path on 7 of the 19 datasets it completed; the GA beats GOSDT on 10. GOSDT did not complete on `qsar_biodeg`, where its search exhausted memory under caps of 8 and 11 GB; that dataset is excluded. It took 14.4 s per fold on average; 32 of 1900 fits hit the 30 s limit and are therefore not certified optimal, and 10 returned no model and were dropped from the path (which can only lower GOSDT’s hypervolume). An exact sparse-tree solver, in other words, does not dominate CART’s pruning path on these datasets either, and the difference between the GA and GOSDT is not significant (-0.18 , Holm $p = 0.980$). The pruning path is a harder baseline than a single CART tree suggests. This comparison was added after K1/K2 were known and decides nothing.

6 WHAT THE AUDIT FOUND

The audit behind this study produced five findings that may be of use beyond this system.

Published numbers without runs. The “46–82%” size reduction was spliced from two dictionaries of literals in a plotting script, and a figure titled “Statistical Equivalence to CART” plotted cross-fold t -test p -values against a hand-set target of 0.55. The fix was structural rather than editorial: the figure generator now reads only committed run output and raises an error when there is none, and every number in this paper is generated from committed CSVs.

Selection on the test fold. The first frontier run reported K1 triggered, with RS ahead by a wide margin on 19 of 20 datasets. RS had been returning every tree it sampled—about 2,500 per fold—and the dominance filter then ran on their test scores: a maximum over thousands of test evaluations that no method could deliver, since choosing among the candidates needs the test labels. The error was visible in the delivered count, 2,545 against the GA’s 10. With both searchers archiving on validation objectives, they deliver 10.2 (GA) and 5.6 (RS) models per fold and the verdict reverses.

The reference point. The pre-registration fixes the hypervolume reference per dataset; the harness computed it per fold. A smaller box favours whichever method has the lower peak accuracy—here the GA—and the per-fold version put the share of datasets on which the GA has the larger hypervolume at 70%, above the K2 threshold. Computed as specified, it is 45%. This was found by reading the implementation against the protocol, not prompted by the result, but it illustrates why a pre-registration must fix the metric’s parameters and not only its name.

A saved configuration that does not reproduce its run. Each run saved its resolved configuration next to its output. The YAML writer sorted keys alphabetically, and the mutation operator is drawn by position from the ordered weight table, so re-running from the saved file selected different operators for the same seeds and produced different fronts. We found this only because an ablation’s control arm failed to match the committed run; the source configuration reproduces it exactly. Serialisation order is part of the experiment whenever a random draw indexes into a mapping.

Diagnoses that were wrong. Two pre-run screening measurements (three datasets, three folds each, so directional only) contradicted our prior explanations. Midpoint thresholds were adopted to remove a handicap we believed was suppressing the GA; they raised *random search*’s accuracy by 0.012 and lowered the GA’s by 0.003. A plausible explanation is that the GA could already reach good thresholds through threshold mutation over generations, whereas RS, which draws every candidate independently, could not. Lowering the initial stopping probability to 0 to seed bushier trees made the GA lose to RS (-0.018): seeded with full-depth trees, RS keeps the large accurate ones (16.7 leaves) while the GA, whose mutations favour pruning five to one and whose single-objective fitness rewards small size, ends at 4.4 leaves. Both decisions—keep midpoint thresholds, keep the stopping probability at 0.3—were recorded, with the direction they push K1, before the frontier run.

7 DISCUSSION AND LIMITATIONS

The positive result is narrow: over one representation, one fitness and one budget, NSGA-II outperforms random sampling of the same space with a medium effect size ($d_z = +0.50$). This supports a claim about the search mechanism, not a claim about predictive performance relative to standard methods. The negative result is also narrow. The GA was not tuned for the frontier run. The ablation suggests that a larger budget and fitness without a small validation split each recover part of the gap, and greedy seeding with CART trees—which would place the large accurate trees in the population directly—is untested. We did not pursue any of them, because tuning the GA after observing K2 would introduce the analytic flexibility the pre-registration is meant to prevent; they are hypotheses for a new pre-registered study. The datasets are mid-sized (500–1458 rows, up to 41 features) by design; the K3 run used one outer repeat; the hypervolume reference depends on the largest tree any method produced; and nothing here measures whether a human can read these trees faster—size proxies are not interpretability [13, 22].

For the evaluation of evolutionary tree induction we make three recommendations. Report budget-matched random search over the same representation; it is cheap, and without it no advantage can be attributed to evolution. Compare frontiers against the full cost-complexity path, not a single CART tree. And fix the metric’s parameters in the protocol, including reference points and the rule for which models a method delivers: both of our harness errors lived there, and each was large enough to reverse a conclusion.

8 CONCLUSION

In this system, evolutionary search outperforms budget-matched random sampling of the same tree space but not CART's cost-complexity pruning path, which requires no search over whole trees. We report this as the main finding, together with the audit trail—the withdrawn claims, the two reversed verdicts and the committed data behind every number—so that the next comparison in this area can start from the baselines that decided this one.

AVAILABILITY

Code, configuration, pre-registration, deviation log and all run output are in the project repository (anonymised for review).

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